

AI Core Logic & Biomechanics Report

To: AI Development Team

1. Core Tracking Logic (Vector Math)

We implemented a robust joint tracking system based on 3 primary landmarks: Hip, Knee, and Ankle. The C# script uses `Vector3.Angle` to calculate the interior angle between two vectors (Knee-to-Hip and Knee-to-Ankle) frame by frame.

2. Repetition Validation (State Machine)

To guarantee accurate counting, a rep is only registered if it satisfies a strict boolean state machine:

- **inProgress:** Triggered when the patient leaves the resting angle.
- **hitGreenZone:** Triggered when the patient enters the medically approved target angle.
- **hitRedZone:** Triggered if the patient exceeds the safe range (bad form).
- **Validation:** A count is strictly awarded if the patient hits the Green Zone, avoids the Red Zone entirely, and returns to the resting position.

3. Medical Biomechanics & Angle Justifications

Exercise A: Squat

- Rest Angle: > 140 degrees (Standing).
- Target Zone (Green): 70 to 100 degrees. This optimally activates the Quadriceps and Glutes while minimizing shear stress on the ACL.
- Danger Zone (Red): < 70 degrees. Dropping below this angle significantly increases stress on the patellofemoral joint.

Exercise B: Seated Leg Extension

- Rest Angle: < 110 degrees (Seated).
- Target Zone (Green): 150 to 180 degrees. The clinical goal is to achieve Terminal Knee Extension (TKE) to fully engage the VMO muscle.
- Danger Zone (Red): > 180 degrees. Pushing beyond 180 induces knee hyperextension, risking ligament damage.

4. Scalability

The logic is fully modular. Adding a new physical therapy exercise only requires defining its name in the `ExerciseType` enum and setting its Min/Max/Rest angles. The core math engine handles the rest automatically.